LAB TASK Date : 06-10-2025

Problem Statement 1: Array Operations in Daily Life

***Scenario*:**
You are working as a data analyst for a sports team. The coach wants to analyze the players' scores from recent matches. You are given a list of player scores, and your task is to manage and analyze the data using NumPy arrays.

***Task*:**
Write a Python program that:

1. Creates a NumPy array with the players' scores.
2. Adds a new score to the array.
3. Inserts a score at a specific position in the array.
4. Removes a particular score from the array.
5. Finds the position of a given score.
6. Sorts the scores in ascending order.
7. Reverses the order of the array to show the top scores first.

***Example*:**
If the scores are `[45, 60, 72, 51, 89]`, demonstrate each operation step by step.

Problem Statement 2: Smart Calculator for Daily Use

***Scenario*:**
You are building a simple calculator app for your friend who is learning math. The calculator should help perform basic operations and also handle both real and complex numbers for more advanced calculations.

***Task*:**
Write a Python program that:

1. Defines separate functions for addition, subtraction, multiplication, and division.
2. Uses the **math** module to perform operations like finding the ceiling value of a number, exponential value, and greatest common divisor.
3. Uses the **cmath** module to find the tangent, logarithm base 10, and to check if two complex numbers are approximately equal.
4. Allows the user to input numbers and select which operation to perform.

***Example*:**
If the user inputs two numbers like 5 and 3, the calculator should display results for all four operations. If complex numbers are given, it should use cmath functions.